Polynomials – 1

Roots, multiplicity and the derivative

Homework

8 сентября 2026 года

Theoretical minimum. Multiplicity of a root and the derivatives vanishing there; $\gcd(f, f')$ as the multiple part; Rolle's theorem for polynomials and the interlacing of the roots of f and f'; Vieta's formulas; a non-negative real polynomial is a sum of two squares; the sign of a polynomial between consecutive roots.

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1. (Putnam 1948) Let $f(x)$ be a cubic polynomial with roots x_1, x_2, x_3 and assume that $f(2x)$ is divisible by $f'(x)$. Compute the ratios of the roots.

2. (Putnam 1953) Assuming that the roots of $x^3 + px^2 + qx + r = 0$ are all real and positive, find the relation between p, q and r which is necessary and sufficient for the roots to be the cosines of the angles of a triangle.

3. (Putnam 1953) Show that the roots of $x^4 + ax^3 + bx^2 + cx + d = 0$, suitably numbered, satisfy $r_1/r_2 = r_3/r_4$ provided that $a^2d = c^2 \neq 0$.

4. (Putnam 1958) If $a_0, a_1, \ldots, a_n$ are real numbers with $\sum_{k=0}^{n} \frac{a_k}{k+1} = 0$, show that the equation $a_0 + a_1 x + \cdots + a_n x^n = 0$ has at least one real root.

5. (Putnam 1978) The polynomials $Q_n(x)$ are defined by $Q_1(x) = 1 + x$, $Q_2(x) = 1 + 2x$ and, for $m \geqslant 1$, by $Q_{2m+1} = Q_{2m} + (m+1)xQ_{2m-1}$ and $Q_{2m+2} = Q_{2m+1} + (m+1)xQ_{2m}$. Let x_n be the largest real solution of $Q_n(x) = 0$. Prove that (x_n) is increasing and that $x_n \to 0$.

6. (Putnam 1981) Let $P(x)$ be a real polynomial and put $Q(x) = (x^2 + 1)P(x)P'(x) + x\big((P(x))^2 + (P'(x))^2\big)$. Given that $P(x) = 0$ has n

distinct real roots exceeding 1, prove or disprove that $Q(x) = 0$ has at least $2n - 1$ distinct real roots.

7. (Putnam 1990) Does there exist an infinite sequence $a_0, a_1, a_2, \ldots$ of non-zero real numbers such that for every n the polynomial $p_n(x) = a_0 + a_1 x + \cdots + a_n x^n$ has exactly n distinct real roots?

8. (Putnam 1941) Prove that the polynomial $(a-x)^6 - 3a(a-x)^5 + \frac{5}{2}a^2(a-x)^4 - \frac{1}{2}a^4(a-x)^2$ takes only negative values for $0 < x < a$.

9. (Putnam 2003) Suppose that a, b, c, A, B, C are real numbers with $a \neq 0$ and $A \neq 0$ such that $|ax^2 + bx + c| \leqslant |Ax^2 + Bx + C|$ for all real x. Show that $|b^2 - 4ac| \leqslant |B^2 - 4AC|$.

10. (Putnam 2024) For which real polynomials p is there a real polynomial q such that

$$p(p(x)) - x = \big(p(x) - x\big)^2 q(x)$$

for all real x?